



A Systematic Review Mapping of AI Literacy Progression in K–12

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Abstract

Several artificial intelligence (AI) literacy frameworks have been developed to guide what K–12 education should focus on to prepare students to be AI-literate. However, these frameworks tend to either focus heavily on technical content or fail to specify the competencies students need at each grade band to be considered AI-literate. To address this gap, this study conducted a systematic review to map AI literacy competencies, examining the focus of K–12 AI literacy and education research, the specific competencies emphasized at each grade band, and how these competencies evolve across grade bands. We reviewed 36 empirical studies and identified five components of AI literacy that have been studied so far: Understanding Foundational AI Concepts, Creating AI Artifacts, Interacting With AI Agents, Developing Awareness of AI Ethics, and Understanding Human-AI Relationships. Additionally, we found evidence of how specific AI literacy competencies associated with each of these components progress and evolve across grade bands. Our findings highlight gaps in the existing research on AI literacy in K–12 education and offer insights into the practical implications for developing AI literacy curricula that are age-appropriate and comprehensive.

Keywords AI literacy · Progression · K–12 · Systematic review

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Introduction

Artificial intelligence (AI) and AI-based technologies are becoming increasingly prevalent in every aspect of our daily lives, significantly transforming sectors such as education, business, medicine, finance, and the workforce (National Science and Technology Council, 2016). These technologies are described as intelligent systems and algorithms that can reason and adapt based on established rules and environmental inputs, thereby emulating human intelligence (McCarthy, 2007). Consequently, they are capable of performing cognitive tasks, including learning and problem-solving (Zawacki-Richter et al., 2019). In education, for example, AI technologies, such as natural language processing tools, have been shown to enhance students' learning experiences by supporting epistemic process skills, including argumentation, scientific explanation, and critical thinking (Chng et al., 2023).

The rapid advancement of AI technologies has prompted many countries to prioritize AI education, especially at the K–12 level, to ensure that people are well informed and literate about AI (Chen et al., 2023). Public discourse reflects this urgency: a large-scale sentiment analysis of social media posts revealed that conversations about STEM education and AI often emphasize coding, data science, and technology, which highlights the need to meaningfully connect AI to STEM education (Smith-Mutegi et al., 2025). Many argue that AI literacy is essential for students to learn in K–12 education (Ottenbreit-Leftwich et al., 2023), and various initiatives and frameworks have been developed to define it. Pinski et al. (2024) argue that AI literacy differs from technology literacy because of the unique characteristics of AI, such as autonomy, learning capability, and inscrutability, which challenge common assumptions about human–technology interactions. These characteristics further emphasize the need for people to become AI-literate so they understand and respond effectively to this paradigm shift in human–technology interactions.

Developing a framework is imperative for nurturing AI literacy, enabling curriculum researchers and developers to design necessary and targeted interventions that help individuals become AI-literate. Among the first AI literacy frameworks developed are the “Five Big Ideas in AI” suggested by Touretzky et al. (2019), the competencies within AI literacy by Long and Magerko (2020), and the AI literacy cognitive domains by Ng et al. (2021a). Recently, others have developed slightly different but related AI literacy frameworks (e.g., Chiu et al., 2024; Mills et al., 2024). Although there are differences in these frameworks, such as referring to AI literacy as comprising “literacy,” “components,” or “competencies,” they generally share a common understanding. AI literacy is widely interpreted as an individual's confidence and capability to clearly explain how AI technologies function and affect society, as well as to use those technologies ethically and responsibly and to communicate and collaborate effectively with them in any context (Chiu et al., 2024). In response to the increasing number of AI literacy and education frameworks, Casal-Otero et al. (2023) conducted a systematic literature review on AI literacy in K–12 and found that the frameworks do not clearly break down the components and competencies of AI literacy by grade bands. This makes it challenging for researchers and curriculum developers to operationalize and concretize them in the

curriculum. The only framework that has been unpacked and broken down by grade band is the Five Big Ideas in AI (Touretzky et al., 2019), which outlines learning objectives for four grade bands: K–2 (lower elementary), 3–5 (upper elementary), 6–8 (middle school), and 9–12 (high school). Therefore, this framework has been useful for teachers and curriculum developers. However, this type of framework, often called a “learning objective progression,” focuses more on the cognitive aspect of AI literacy, such as understanding AI technical concepts, and underemphasizes the noncognitive aspects of AI literacy, such as awareness and self-efficacy of using and learning about AI. Recent research supports this concern: Usher and Barak (2024) found that online explicit–reflective learning modules can strengthen STEM students’ ethical knowledge, awareness, and problem-solving skills, demonstrating the importance of noncognitive competencies in AI literacy education.

We were motivated to conduct a systematic literature review to examine which AI literacy components and competencies have been studied at each grade band and to determine whether an AI literacy progression exists in the noncognitive components of AI literacy. Such a study is crucial not only for understanding the progression of AI literacy components addressed at each grade band and how they develop over time across grades, but also for identifying gaps in research on these components within the context of K–12 AI literacy and education. With these goals in mind, our research questions were as follows:

1. What AI literacy components have been studied in K–12 education?
2. How do the competencies within each AI literacy component evolve across different grade bands in K–12 education?

Related Work

AI Literacy Frameworks in K–12 Settings

Scholars have proposed many AI literacy and education frameworks, and one of the most widely used and popular frameworks is the Five Big Ideas in AI (Touretzky et al., 2019). This framework was initially intended to familiarize students, teachers, and the general public with the core technical concepts and major issues of AI, which are often misunderstood and confused with science fiction. The Five Big Ideas are Perception, Representation and Reasoning, Learning, Natural Interaction, and Societal Impact.

Perception involves understanding how computers perceive the world using sensors, defined as the “extraction of meaning from sensory signals, using knowledge” (Touretzky & Gardner-McCune, 2022, p. 155). For example, in speech recognition, computers rely on linguistic knowledge, such as sentence structure, morphology, and semantics. In visual perception, computers derive meaning from image components such as pixels, edges, and contours. This concept is often introduced in discussions about the challenges AI technologies, such as Siri and Alexa, face in recognizing diverse accents and speech patterns. *Representation*

and Reasoning, the second Big Idea, explores how data are structured and how algorithms perform reasoning. This idea is closely tied to computer science and is often taught through activities like creating decision trees for guessing games involving objects, animals, or people. *Learning*, the third Big Idea, focuses on how computers use data to find patterns and optimize approaches through specialized algorithms. Unlike human learning, which is general and flexible, machine learning is task-specific, relying on algorithms to perform particular functions. The next Big Idea, *Natural Interaction*, includes topics such as natural language understanding, common sense reasoning, affective computing, and theory of mind. These areas enable AI to interact with humans by leveraging the various forms of knowledge it has been trained on. Lastly, *Societal Impact* examines the positive and negative effects of AI on human life. This Big Idea encompasses four subtopics: the ethics of AI decision-making, the economic impacts of AI technology, the relationship between AI and culture, and the use of AI for social good. It also highlights the biases in AI rooted in the data used for training and the algorithms developed.

Touretzky et al. (2019) further developed these Big Ideas into detailed concepts and created learning progressions, outlining how each concept can be progressively introduced to students from K–12, 3–5, 6–8 through grades 9–12. While the Five Big Ideas framework is an essential and valuable contribution, many scholars argue that it focuses too heavily on the technical aspects of AI literacy, despite the inclusion of Societal Impact as a Big Idea. As a result, alternative frameworks have emerged, such as the recently published AI Learning Priorities for All K–12 students, co-authored by the Computer Science Teacher Association (CSTA) and AI4K12 (2025), which offers an updated version of the Five Big Ideas in response to the rapid advancements in AI technologies, including Generative AI tools. Grounded in these revised ideas, the report identifies five core learning priorities for K–12 students: Humans and AI, Representation and Reasoning, Machine Learning, Ethical AI System Design and Programming, and the Societal Impact of AI. These priorities are intended to guide K–12 learning outcomes, inform curriculum design, and support educators in developing effective pedagogical approaches for AI education (CSTA & AI4K12, (2025)). Others, such as those proposed by Long and Magerko (2020), Ng et al. (2021), and UNESCO’s AI literacy framework (Miao & Shiohira, 2024). These frameworks aim to provide a more balanced approach by incorporating both the technical and nontechnical dimensions of AI literacy and education.

Long and Magerko (2020) conducted a review study aimed at identifying the competencies necessary for individuals with limited or no technical understanding of AI to effectively interact with and evaluate AI systems. While this study did not focus on K–12, their focus on non-technical understanding of AI offers significant insights into how the findings can be translated into AI literacy in K–12 settings. They proposed their own definition of AI literacy as “a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace” (p. 2). Based on their findings, the authors proposed 17 competencies, grouped into four categories: What Can AI Do? How Does AI Work? How Should AI Be Used? and How Do People Perceive AI?

What Can AI Do? includes six competencies: distinguishing technologies that use AI from those that do not; understanding what makes a technology “intelligent”; recognizing the interdisciplinary nature of AI; differentiating between general and narrow AI; identifying the types of problems AI excels at solving and those where it fails; and predicting possible future applications and the effects of AI on society. The second category, *How Does AI Work?* focuses on nine technical competencies: understanding knowledge representations; understanding how machines reason and make decisions; identifying the steps involved in machine learning; recognizing the role of humans in developing and fine-tuning AI systems; understanding data science concepts and practices; learning how computers use data to learn; understanding the role of sensors in AI systems; recognizing how some AI systems can physically interact with the world; and understanding how higher-level reasoning directs AI systems in physical actions. *How Should AI Be Used?* includes one competency: identifying and describing various perspectives on key ethical issues related to AI, such as privacy, employment, misinformation, the singularity, ethical decision-making, diversity, bias, transparency, and accountability. Finally, *How Do People Perceive AI?* also includes one competency: understanding that AI agents are programmable. This framework highlights a comprehensive approach to AI literacy by addressing both technical and ethical dimensions, equipping individuals to critically engage with AI technologies in various contexts.

Grounded on the 17 competencies of AI literacy, Long and Magerko (2020) further proposed 16 design considerations to inform AI developers and educators as they design and develop learner-centered AI experiences and interventions. Noticeably, they synthesized existing research to highlight the approaches that can be adapted to support early exposure to AI technology in shaping children’s perceptions of AI and cultivating foundational AI literacy skills.

Ng et al. (2021) conducted a review study and identified four overarching components of AI literacy that align with Bloom’s taxonomy. The first component, *Knowing and Understanding AI*, involves understanding basic AI functions and knowing how to use AI applications. The second, *Using and Applying AI Knowledge*, entails applying AI concepts and knowledge to different scenarios. Third, *Evaluating and Creating AI Applications* includes critically analyzing and designing AI applications. Finally, *AI Ethics* focuses on human-centered considerations, emphasizing ethical implications.

The most recent framework, developed by UNESCO, defines AI literacy as the competencies required to understand how AI collects, cleans, manipulates, and analyzes data, as well as algorithm literacy, which refers to the ability to understand how AI algorithms identify patterns and connections in data, often used for human–machine interactions. Based on this definition, UNESCO created two frameworks: one for teachers (Miao & Cukurova, 2024) and one for students (Miao & Shiohira, 2024). The student framework outlines 12 competencies across four dimensions: *A Human-Centered Mindset*, *Ethics of AI*, *AI Techniques and Applications*, and *AI System Design*. In addition to defining these components, the authors demonstrate how each competency progresses from *understanding* to *application* to *creation*. The embedded progression levels (understand, apply, and create) were designed to scaffold students’ development toward increased “sophistication,

proficiency, and ethical consciousness in using and co-creating AI technology” (Miao & Shiohira, 2024, p.20).

At the *Understand* level, which is designed for all students as they will increasingly engage with AI in various aspects of their life, students are expected to develop a foundational understanding of AI concepts and practices. This includes developing age-appropriate insights into the values, ethical issues, core principles, processes, and methods that underpin AI systems and applications. Learners should be able to demonstrate their understanding by relating concepts to real-world contexts and by integrating new information into their existing knowledge base. The *Understand* level is considered a critical starting point to foster students’ perspectives around AI to embrace more learning opportunities in their future study. The *Apply* level is designed to engage all students while offering educators the flexibility to differentiate instruction by adjusting the scope, depth, and complexity of learning objectives or thematic content. At this stage, students are expected to extend their understanding and adapt their knowledge, values, and skills to novel contexts. This includes applying a human-centered and ethical mindset to evaluating, exploring, and using AI tools in practical settings. Lastly, the *Create* level expects students to take on the role of conscientious AI co-creators. This includes designing human-centered solutions that address real-world needs and promoting responsible development and use of AI. More critically, students at this level are encouraged to apply their understanding of data, algorithms, and ethical design in the development of AI applications and to engage with ongoing discussion about AI governance and regulation thoughtfully.

While the UNESCO’s AI literacy framework (Miao & Shiohira, 2024) and the Ng et al. (2021) frameworks are aligned with broader conceptualizations of AI literacy, they are explicitly focused on the cognitive and affective domains of AI literacy. Both frameworks offer comprehensive coverage of AI literacy competencies across K–12 education, serving as standard-setting tools for researchers and curriculum designers to inspire the development of AI literacy and education interventions and articulate measurable learning objectives tailored to age-appropriate educational settings. In comparison to these AI-literacy-specific frameworks, some frameworks target specific subdomains of AI. For example, Lao (2020) proposed a Machine Learning Education Framework that organizes learning into three domains: Knowledge, Skills, and Attitudes. Lao used this framework to assess existing machine learning (ML) education, including those in K–12, and concluded that more general and accessible ML education is essential. According to Lao, students need to understand ML technologies, critically interpret ML outputs, and feel confident in participating in ML-related discussions and advocacy within their communities. These objectives are all closely aligned with the broader goals of AI literacy.

All of the frameworks above both overlap and have unique focuses, providing valuable guidance on nurturing AI literacy and education in K–12. However, none of these frameworks explicitly detail how each component of AI literacy should be implemented across grade bands or how it evolves from one grade band to the next. A more fine-grained framework is essential to define what it means to be AI-literate at different educational stages, such as preschool, elementary, middle, and high school. Additionally, many of these frameworks use inconsistent terminology to

describe AI literacy and its components. This lack of alignment motivated us to synthesize the existing frameworks and adopt more unified terminology in our review.

Preparing AI-literate Global Citizens at an Early Age

As mentioned above, while existing frameworks offer valuable foundations for defining AI literacy, they often lack specificity regarding how AI literacy competencies should be implemented across different grade levels. To better understand how these competencies are being introduced in K–12 settings, we examine global and national efforts that are actively integrating AI into school curricula. According to UNESCO's, 2022 report, 11 UNESCO member states have developed, endorsed, and implemented AI curricula across diverse educational systems (UNESCO, 2022). The integration of these AI curricula varied by context: some UNESCO member states developed AI as a standalone subject within the national or local curriculum frameworks, while others have adopted interdisciplinary approaches, embedding AI concepts into existing subjects, including information and computing technologies (ICT), science, math, language, social studies, and moral education. While a larger share of these curricula targets middle and high school students, several government-supported programs also address early and later primary education, demonstrating active efforts to introduce AI to younger learners (UNESCO, 2022). Similarly, a benchmarking report highlights how countries, particularly China, have aligned AI education with their national AI strategies. The Ministry of Education in China has adopted policies and guidelines for integrating AI, along with ICT and computational thinking (CT), across primary, secondary, and post-secondary education, aligning with its national goals for the development of its AI sector (Laguarda et al., 2022).

Beyond government initiatives, non-governmental organizations and private sectors have also developed AI curricula aimed at engaging younger students in learning about AI and developing emerging AI literacy. For example, the DAILY Curriculum (MIT Media Lab Personal Robots Group & MIT STEP Lab, n.d.) and AI Curricula for Elementary School (Code.org, n.d.) have been designed for broader implementation of AI learning at grade bands prior to high school levels. These curricula emphasize that AI literacy can begin in the early grades and does not need to be restricted to high school. They aim to help students develop a basic understanding of AI concepts and eventually progress to become literate in AI technologies, techniques, and real-world applications. In fact, AI literacy is inherently interdisciplinary, making it well-suited for integration into existing subjects. For example, AI overlaps with data literacy and algorithmic thinking—both are core elements of computer science and CT (e.g., Heintz, 2022). These connections are also reflected in the Computer Science Teachers Association (CSTA) K–12 Computer Science Standards in the United States. According to CSTA, students in kindergarten begin by modeling simple processes through step-by-step instructions, such as creating and following basic algorithms. By grades 3–5, they use flowcharts or pseudocode to address complex problems, and by grades 11–12, students are expected to implement AI algorithms to play games or solve real-world challenges (CSTA, 2017).

This progression illustrates the importance of leveraging the synergy between AI, CS, and CT to introduce AI meaningfully starting in the early grades.

The International Society for Technology in Education (ISTE) has developed AI lessons that include tailored learning objectives and activities for both elementary and secondary educators (ISTE, [n.d.](#)). The *Hands-On AI Projects for the Classroom: A Guide for Elementary Teachers* introduces foundational topics, such as data, machine learning, and sensors, through unplugged and hands-on activities that engage young learners in exploring fundamental AI concepts (Black & Brooks-Young, [2024a](#)). In addition, the *Hands-On AI Projects for the Classroom: A Guide for Secondary Teachers* emphasizes hands-on experiences with AI technologies, including conversational agents and decision-making tools, to deepen secondary students' understanding of AI applications (Black & Brooks-Young, [2024b](#)). These resources together underscore the importance of continuity in fostering AI literacy across grade levels. Recognizing the need to equip all learners with AI literacy, a range of initiatives has emerged at local, state, and national levels in the United States (e.g., Chicago Public Schools, [n.d.](#); North Carolina Department of Public Instruction, [2024](#)), reflecting a growing commitment to fostering AI-ready learners from the early grades onward.

In sum, the synergies between AI literacy and other literacies, including CS, CT, and digital literacy, demonstrate the importance of recognizing AI literacy as a foundational competence for all learners. Contemporary perspectives underscore that technical skills alone are insufficient for being AI-literate. Instead, individuals must be able to reason about AI's broader implications, work collaboratively in AI-mediated environments (e.g., Generative AI tools), and adapt to continuous technological changes, all of which align with the transferable professional competencies in today's workforce (World Economic Forum, [2025](#)). Hence, promoting AI literacy from an early age ensures that all students, regardless of background, are equipped to understand, interact with, and critically engage with AI systems that increasingly shape modern life and the workplace. However, more efforts are needed to understand how AI literacy can be effectively taught in response to children's developmental stages in K–12 education. In particular, there is a gap in understanding how theoretical frameworks based on cultural understandings of AI literacy can be effectively translated into the practical design of AI literacy interventions in educational settings (e.g., Yang et al., [2025](#)). To address this gap, we conducted a review study to examine existing research on AI literacy across grade bands and identify the components of AI literacy that have been emphasized at each level. This work not only validates the learning progressions proposed by existing frameworks (e.g., the Five Big Ideas of AI) but also highlights gaps in research that future studies can address.

Methods

Systematic Literature Review

To derive AI literacy components and competencies, we conducted a systematic literature review consisting of four phases following the Preferred Reporting Items for

Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2010). Our process started with the *identification phase*. To identify documents through two platforms, ProQuest and Web of Science, we set the keywords of “AI literacy” OR “Artificial Intelligence literacy” with several word strings (“AI thinking” OR “Artificial Intelligence thinking” OR “AI education” OR “learning about AI” OR “teaching AI” OR “ML literacy” OR “understanding ML” OR “understanding AI” OR “AI for K–12” OR “AI university” OR “AI courses” OR “AI school” OR “AI informal learning”). This phase resulted in 430 articles that matched our keywords. In the next phase, two education researchers *screened* the titles and abstracts of the identified articles based on our inclusion and exclusion criteria, as shown in Table 1.

We then examined the full texts of this subset of articles to further determine their *eligibility* based on those inclusion and exclusion criteria. While examining those eligible articles, we also used the snowball approach to identify additional relevant sources from among those cited in the articles. These efforts resulted in 36 articles included in our final literature review sample as shown in Fig. 1. Finally, we *synthesized* key findings from these 36 articles, separated by grade bands from elementary to high school, through a thematic analysis (Saldaña, 2021). We focused on gathering AI literacy competencies and ways that AI is integrated into STEM or science disciplines. Two researchers reviewed all the articles and extracted evidence. The researchers thoroughly examined all discrepancies arising during the review process and discussed them until they reached a consensus.

Analytical Approach

While several AI literacy frameworks exist, as described in the Related Work section above, we decided not to use any specific framework to capture nuanced AI literacy components that might not have been included in or addressed by the existing frameworks. Instead, we employed a qualitative coding approach using the constant comparative method (Strauss & Corbin, 1998). This began with an open coding process (Saldaña, 2021), in which two experienced education researchers (the first and second authors of this paper) with backgrounds in STEM and CS education identified

Table 1 Inclusion and exclusion criteria for literature review

Inclusion criteria	Exclusion criteria
Mentions AI education and AI literacy	Does not mention AI education or AI literacy
Primary focuses on students and student learning	Primary focuses on teachers and teacher education, or professional development programs
Situated in kindergarten to high school (K–12) grades	Situated in post-secondary settings
Empirical study or systematic review studies published in peer-reviewed journals	Reports, briefs, editorials, or non-peer-reviewed publications
Studies published between January 2018 and March 2024	Studies published before 2018
Written in English	Written in other language than English for the main text

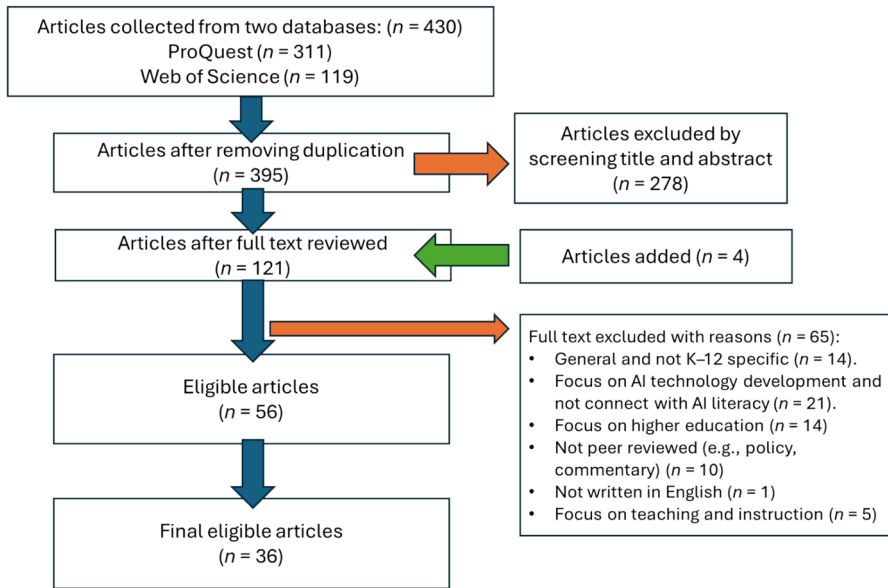


Fig. 1 PRISMA flow chart for the literature article collection and review process (Moher et al., 2010)

and collected AI literacy categories (e.g., big ideas) and competencies (e.g., learning objectives) reported in the empirical studies of the reviewed articles. Additionally, we gathered grade-level information. Given that grade band configurations vary across jurisdictions and countries, we carefully recorded the specific grade levels and/or age ranges referenced in each study. Based on this information, we established the following grade band groupings: early elementary (K–3, ages 5–8), upper elementary (4–6, ages 9–11), middle school (7–8, ages 12–13), and high school (9–12, ages 14–18). This segmentation allows us to account for developmental differences across the elementary years and better align our analysis with international educational structures. Prior to open coding, the two coders each reviewed a random subset of the articles and then convened to discuss examples that could be included as categories and competencies, developing a shared understanding of the open coding criteria. The open coding primarily focuses on the content described in the studied programs (e.g., interventions, curricula), including learning objectives, AI concepts, AI practices, and attitudinal perspectives related to AI. Moreover, the coding process also captures how AI literacy was being defined, measured, and reported in these studies. Each coder was then randomly assigned half of the included articles to review and extract relevant information. The coders met regularly to discuss emerging questions and ensure consistency. The first coder also reviewed all the papers to ensure that all relevant categories and competencies were included.

After completing the open coding process, we conducted axial coding (Saldaña, 2021), grouping the categories and competencies into higher level components based on their similarities and differences in terms of AI literacy concepts and practices. However, we retained the categories and competencies

identified during the open coding process as sub-components or sub-competencies. For this process, the same coders each took the categories and competencies for the K–6 grade levels captured from the open coding, then summarized these into different higher-level components. The coders then met and compared their summaries and discussed the components and resolved any discrepancies. All the consensus meetings resulted in a narrow set of components, along with their definitions, example categories, and competencies, and then modified the categories that can be mapped into the other grade bands. The first authors then applied these components and definitions to the remaining open codes across the middle school and high school levels. We reported these higher level emerging components to address the first research question regarding the AI literacy components studied in K–12 education.

To answer the second research question about how each component evolves across different grade bands, we compared the competencies (e.g., sub-competencies and/or sub-components) within each broader component. For this process, the same two coders each reviewed the sub-competencies and sub-components for the elementary grades and then broke down these codes into K–3 and 4–6. Each of the coders wrote the summaries of the competencies that belong to lower or upper primary elementary levels, then compared both coders' summaries and finalized the comparison results. The first author then applied this process to summarize the competencies in other grade bands. This comparison allowed us to describe the specific focus of AI literacy at particular grade bands. The two researchers conducted the review and coding processes collaboratively and engaged in rigorous qualitative discussions to minimize potential biases. They discussed and resolved any discrepancies in coding and interpretation through consensus.

Limitations

We acknowledge that while this paper offers a comprehensive analysis of AI literacy progression in K–12 education based on a systematic review of the literature, several limitations exist that warrant more careful interpretation of findings. First, the data collection relied primarily on two databases, ProQuest and Web of Science, and covered publications from 2019 to March 2024. As a result, the study may not capture relevant literature indexed in other databases such as SCOPUS or Google Scholar, and does not include studies published after March 2024. Second, our research strategy was intentionally focused on the term of AI literacy, rather than encompassing broader related competencies or adjacent domains (e.g., computing literacy). This narrow scope may have excluded studies that address important aspects of AI literacy without explicitly labeling them as such. Finally, regarding the analytical approach, given the nature of the open coding process, we did not conduct a reliability consensus. However, we documented our discussion process and decision points, and the first author performed consistent checks on the coding process to ensure content accuracy.

Results

RQ1. What AI Literacy Components Have Been Studied in K–12 Education?

Based on our review findings, we identified five components of AI literacy that have been the focus of empirical studies in K–12 education. These components are (1) Understanding Foundational AI Concepts, (2) Creating AI Artifacts, (3) Interacting With AI Agents, (4) Developing Awareness of AI Ethics, and (5) Understanding Human-AI Relationships. Figure 2 describes the counts of the reviewed papers addressing each AI literacy component by grade band.

As shown in Fig. 2 (full details in Appendix A), we found that *Understanding Foundational AI Concepts* was the most studied competency across the three grand bands. We also found that *Creating AI Artifacts* and *Interacting With AI* were the main focus at the elementary school level (grades K–6), with a primary focus on cultivating young students' awareness of and interest in AI. At the middle school level (grades 7–8) and high school level (grades 9–12), the emphasis switched particularly to *Creating AI Artifacts*, *Developing Awareness of AI Ethics*, and *Understanding Human-AI Relationships*. Specifically, middle school students were provided with opportunities to create AI-based solutions to address personal or community-relevant issues, such as developing AI-driven solutions for their schools. High school students were often engaged in project-based learning situated in professional settings, such as analyzing customers' reviews for a local store. Throughout the process of creating AI-driven systems and artifacts, middle and high school students would establish an understanding of how different steps within the process could involve bias and other ethical considerations. Such understandings are critical for students to develop in-depth perceptions about human-AI relationships and collaboration in creating positive societal impacts (Fig. 3).

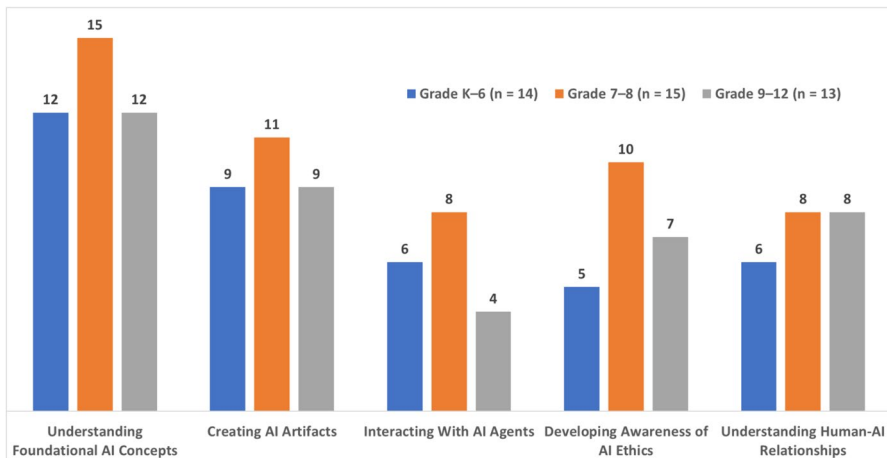


Fig. 2 Counts of the reviewed papers addressing the AI literacy components by grade band

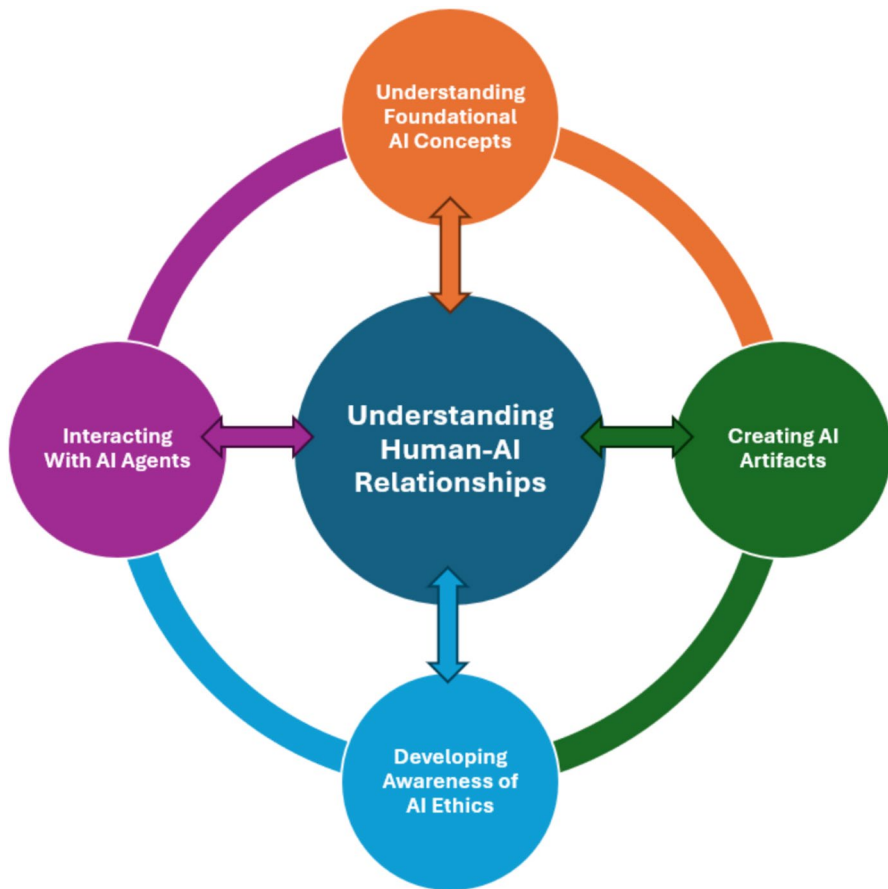


Fig. 3 Relationships among the five AI literacy components

RQ2. How do the Competencies Within Each AI Literacy Component Evolve Across Different Grade Bands in K–12 Education?

Table 2 illustrates the progression of each AI literacy component from elementary to high school. It highlights how the competencies within each component evolve in complexity and focus across grade bands.

Component 1: Understanding Foundational AI Concepts

Based on our findings, studies on AI literacy in *elementary school* have primarily focused on competencies on basic AI concepts as part of foundational AI literacy. Students in elementary grades may have early exposure to various AI technologies. However, the types of technologies that students have experienced vary by factors

Table 2 Learning progression of ai literacy competencies in k–12 education, by grade band

Component	Competency specification K–6	Competency specification 7–8	Competency specification 9–12
1. Understanding Foundational AI Concepts	A. Understand the working principles behind AI. B. Recognize AI in various forms (e.g., Alexa home assistant, Jibo, and Cozmo robots) in everyday life. C. Distinguish AI and non-AI technologies, including automation and robotics. D. Understand basic AI functions and intelligence, including how AI works through gathering, processing, and making decisions based on data. E. Demonstrate basic AI understanding of the concepts of image and voice recognition, content recommendation, and machine learning. F. Understand that computers need to be taught how to perform certain tasks.	A. Define AI and articulate what makes a system or application “intelligent.” B. Explain key AI concepts, including rule-based systems; the Turing test; perception, reasoning, and representation (e.g., pixels and color in computer vision); neural networks; generative adversarial networks (GANs); and natural language processing. C. Identify and explain real-world AI applications in daily life, highlighting their impact and functionality. D. Integrate cultural and humanistic perspectives when understanding and discussing AI, recognizing its societal implications.	A. Understand the core principles of AI, including key basics such as perception, action, representation, reasoning, learning, and artificial collective intelligence, along with critical subfields of AI such as machine learning. B. Recognize the functional modules within AI systems, such as the application of infrared sensors, timers, facial recognition technology, and robotics, which are essential to how AI operates in real-world scenarios. C. Understand the processes that constitute the AI ecosystem, focusing on the internal mechanisms of AI agents and how they achieve intelligent behaviors. D. Understand the history of AI and its evolution, and the key milestones that have shaped the field.

Table 2 (continued)

Component	Competency specification K-6	Competency specification 7-8	Competency specification 9-12
2. Creating AI Artifacts	A. Design AI artifacts through various approaches (e.g., drawing, verbal expression). B. Program simple AI agents to train data with some commercial coding applications (e.g., Teachable Machine). C. Understand that AI requires a comprehensive dataset for training. D. Experience how machine learning functions through direct interaction. E. Understand that AI should require training with a good amount of quality data. F. Explain or illustrate how AI works, plans, or makes decisions and what its expected outcomes are.	A. Design and prototype AI systems that align with personal interests, focusing on logical reasoning and the interaction between human and computer perceptions (e.g., emotion recognition, language processing). B. Utilize both block-based and text-based programming languages (e.g., Scratch, Python) to create functional AI artifacts, such as robots and smart devices, that address real-world challenges. C. Build and train basic machine learning models, incorporating sensor data (e.g., from cameras), and engage in the full development cycle, including data identification, labeling, and model training. D. Develop AI applications with creativity, applying innovative approaches to solve problems or enhance user experiences. E. Integrate knowledge of both hardware and software components in the development of AI applications, ensuring a holistic understanding of AI systems.	A. Identify real-world problems that can be addressed using AI-driven solutions. B. Design and prototype (e.g., applications, hardware, conceptual proposals) and use tools such as Teachable Machine for creating AI systems. C. Develop AI systems by building machine learning models with various data types (e.g., text, images) to solve problems and implement these solutions on real-world devices, such as developing robots for tasks like speech evaluation and real-time recording. D. Program simple AI agents or systems using foundational programming and mathematical skills, utilizing platforms such as App Inventor, Scratch, and Python to construct functional AI artifacts. E. Debug and optimize AI projects to refine and enhance the performance of AI systems, ensuring that they operate effectively and meet the intended objectives. F. Incorporate human insights into the development and testing phases of AI models, ensuring that the artifacts created align with human needs and ethical considerations.

Table 2 (continued)

Component	Competency specification K-6	Competency specification 7-8	Competency specification 9-12
3. Interacting With AI Agents	A. Understand the basis of how an AI agent can learn from data, which is through classification. B. Understand distinctive features of data and representations (e.g., images). C. Understand how robots perceive the world through sensors. D. Interact (play and talk) with AI agents (e.g., asking questions, using robots for creative tasks). E. Observe AI agents' behaviors. F. Use AI products in various contexts (online, at school).	A. Understand how AI-related agents work, such as smart home systems, chatbots, and robots. B. Identify AI applications integrated into daily life, gaining practical experience with these technologies. C. Utilize AI systems in a human-centered manner, ensuring that AI serves human needs and reflects human values.	A. Develop a deeper understanding of AI by directly interacting with technologies such as facial recognition, allowing for experiential learning about how AI processes and responds to inputs. B. Apply AI-based software, such as intelligent writing assistants, to enhance individual learning experiences and demonstrate practical applications of AI in everyday tasks. C. Explore and evaluate how different AI technologies are designed to address specific challenges, gaining insight into the strengths and limitations of AI in solving various types of problems.
4. Developing Awareness of AI Ethics	A. Develop awareness of AI ethics, including privacy, reliability, responsibility, safety, and transparency. B. Understand factors that can lead to algorithmic bias and ways to address this (e.g., quality and quantity of the data). C. Assess critically what AI knows and does not know (e.g., how smart and truthful AI is).	A. Connect ethical concerns with the foundational knowledge of AI and how AI operates to understand its ethical implications. B. Understand core AI ethical principles, including reliability, safety, privacy, responsibility, inclusiveness, transparency, awareness, accountability, and adherence to legal standards. C. Engage in discussions about technology equity and the responsible use of AI, considering the societal impacts of AI technologies.	A. Recognize and evaluate how AI projects and solutions address issues of fairness, bias, and ethical considerations, ensuring that AI applications promote equity and justice. B. Understand and articulate the ethical and legal frameworks governing AI systems, including privacy, accountability, and transparency considerations. C. Critically analyze the decision-making processes of AI models, focusing on the implications of biases in data and algorithms and how they influence outcomes in real-world applications.

Table 2 (continued)

Component	Competency specification K–6	Competency specification 7–8	Competency specification 9–12
5. Understanding Human-AI Relationships	A. Recognize the positive aspects and potential of leveraging AI to solve problems and improve people's lives, as well as the negative implications of AI (e.g., privacy concerns). B. Develop AI readiness and self-efficacy for learning and using AI. C. Envision creative approaches to problem-solving with AI. D. Develop an evolving understanding of AI for social good and the concept of algorithmic bias, which relates to equity in AI models and technology access and impact. E. Understand that AI is related to programming and can be controlled by humans.	A. Understand the critical roles humans play in designing and developing AI systems and algorithms, recognizing the human influence embedded within these technologies. B. Understand the broader social implications of AI, including its potential effects on privacy, security, and societal norms. C. Engage in discussions to explore how AI impacts various interest holders and communities, fostering a deeper understanding of its diverse effects. D. Reflect on the influence of AI on the future of work, particularly how creative applications of AI could reshape industries and job markets.	A. Engage in discussions about AI-related events and their implications on society, focusing on how AI can enhance or challenge the quality of life. B. Explore the ethical role of AI for social good: Delve into AI projects and programs aimed at societal benefit, emphasizing how AI can be leveraged for social good and ethical purposes. C. Cultivate a sustained interest in AI technologies by exploring their potential and future developments through continuous learning and engagement with AI. D. Engage in discussions on the sustainability of AI at various societal levels, considering environmental, economic, and social impacts. E. Understand the importance of human decision-making and accountability in AI systems, highlighting the balance between human control and AI assistance.

such as socioeconomic status, resulting in students having different prior knowledge and even alternative conceptions about AI (Druga et al., 2019). To address this gap, the reviewed research explored ways to engage students in active interaction with AI agents and technologies, with an aim to help them establish AI concepts that are foundational for their subsequent success with AI and AI-integrated learning. The six competencies related to foundational AI concepts will enable students to effectively navigate and critically engage with AI technologies in their personal and future professional lives. Students in *grades K–3* are expected to use plain language, drawing, and/or gestures and movements to explain the processes for training the AI models. For instance, students engage in playful activities, such as dance moves, to understand how various data representations, including images, can be processed by computers through classification models (Su & Yang, 2024). They should also be able to identify distinctive data features, such as edges, shapes, and colors, as they test the predictive power of AI with data. Through these interactions, primarily with AI agents, students learn to understand how AI, especially in robots, perceives the world through sensory inputs (e.g., through cameras connected to the graphic computation units; Yang et al., 2024). Students in *grades 4–6* should understand the basic functions of AI, including how AI models process data and make decisions from their knowledge. Through direct experiences with different AI agents and tools in various contexts, students should develop progressing foundational AI and machine learning concepts, resulting in a growing understanding through distinguishing AI and non-AI technologies in their daily lives (Kim et al., 2023).

At the *middle school level*, the focus shifts to increasing students' understanding of the definition of AI, especially helping students correctly articulate what AI is and what it is not. Given the prevalence of AI-driven technologies and AI-generated media in people's daily lives, students in middle school may have already developed certain perceptions of AI (Kim & Lee, 2024; Kim et al., 2023). As pointed out in our reviewed studies, middle school students may assume that AI is synonymous with automation and robotics, that it is an all-powerful problem solver, or that it operates without ethical concerns (Kim et al., 2023). Because students may interact with AI through social media platforms, voice assistants like Alexa and Siri, and other AI-driven tools, they should develop critical thinking skills to distinguish between AI-powered and non-AI technologies (Druga et al., 2019; Hsu et al., 2022). In middle school, students should be able to accurately define AI and articulate what makes a system or application "intelligent" (Eguchi et al., 2021; Huang, 2021; Park & Kwon, 2024; Williams et al., 2023). For instance, students should be able to understand that AI systems should include machine learning to make decisions based on data, instead of being developed based on pre-programmed processes (Kim et al., 2023). Moreover, middle school students should also move from foundational concepts to specific branches of AI (Eguchi et al., 2021). A broad introduction to AI should include understanding rule-based systems, machine learning, supervised and unsupervised learning, and logic-based reasoning (Zhang et al., 2023). Students should be able to understand how AI makes structured decisions using human-defined rules. They could also explore supervised learning, where models are trained using labeled data, and unsupervised learning, where patterns emerge without explicit labels. In addition to traditional AI concepts, students should be introduced to

modern advancements such as neural networks and generative adversarial networks (GANs; Lee et al., 2021; Williams et al., 2023; Zhang et al., 2023). Understanding GANs, which are used in deepfake technology and AI-generated art, allows students to critically assess AI-generated media and the rising ethical challenges. Obtaining an understanding of essential AI branches equips students with the foundational knowledge to navigate the AI-driven world responsibly (Chiu et al., 2022).

By *high school*, the focus evolves further, emphasizing students' understanding of AI basics and particular branches and their applications to real-world scenarios. Obtaining a solid understanding of AI concepts is essential for high school students as they prepare for postsecondary education and an AI-integrated workforce. At this stage, students engage in more comprehensive explorations of AI foundations, strengthening their conceptual knowledge and practical skills to support career preparedness and future professional aspirations (Kong et al., 2023; Sing et al., 2022). High school students should gain a deeper historical perspective on AI, examining its history and evolution alongside further introduction to various AI branches, including computer vision, natural language processing, and robotics (Chiu et al., 2022). This foundational knowledge serves as a basis for students to design and implement AI models (Bellás et al., 2023).

A critical focus in high school AI education is on obtaining an in-depth understanding of machine learning, which includes key processes such as classification, sentiment analysis, and model development (Chiu et al., 2022; Jiang et al., 2023a, 2023b). Through hands-on experiences with training models, analyzing datasets, and fine-tuning algorithms, students learn how AI systems generalize from data and make predictions. Additionally, students should develop a strong comprehension of data representation and its role in AI decision-making, examining how different data formats influence AI outputs (e.g., Zhan et al., 2022). In high school, understanding the concept of visual recognition, machine learning techniques, and programming related to AI is particularly emphasized to equip students with the ability to build, analyze, and refine AI models. These competencies are essential for students to navigate the rapidly evolving landscape of AI-powered careers and industries, ensuring that students can critically engage with AI technologies and contribute meaningfully to their development and ethical deployment.

Component 2: Creating AI Artifacts

In this crucial component, *elementary school* students often learn about AI systems by designing and developing AI-based artifacts. The process of creating AI artifacts blends students' understanding of AI principles with conceptualization of the design process through creative means (e.g., drawing, gestures, verbal expression, or programming). The key competencies are for students to understand the role of data quality and quantity as well as the model (e.g., machine learning) affecting the training process and the predictive power of AI algorithms or systems. For example, students in *grades K–3* should be able to train a simple AI agent using a non-programming (no-code) and game-based platform (e.g., AI for Oceans; Su & Yang, 2024). This training process primarily involves classifying objects to teach the machine learning model to recognize differences and similarities among these

objects without the need for programming environments. Students in *grades 4–6* are expected to use tangible computing tools to create simple AI agents, such as AI garbage cans, using minimal programming knowledge and skills appropriate for their age range, such as Micro:bit (Hu et al., 2023; Kim & Kwon, 2024).

At the *middle school level*, the focus is on increasing students' understanding of utilizing programming and non-programming platforms to create AI-driven solutions that pertain to students' interests and may benefit their community or society more broadly (Kim & Lee, 2024). By working on projects that align with their interests, students cultivate AI literacy in a meaningful and personal way (Park & Kwon, 2024; Williams et al., 2023). For example, students could design school-based applications that integrate AI to address everyday challenges, such as an AI tutoring app that helps with homework (Kim et al., 2023). Other creative applications include an AI-powered mousetrap, a club recommendation system, or interactive multimedia experiences using AI-powered block-based coding tools (Williams et al., 2023). Students could also explore how AI models learn from data through interactive activities, such as playing the Quick, Draw! game to analyze how image features influence AI classification (Williams et al., 2023), as well as how bias can be embedded in AI systems through dataset selection and model training (Druga et al., 2019). Additionally, students could experiment with facial landmark and hand gesture detection to control virtual environments, which can deepen their understanding of AI application design (Aung et al., 2022). For example, students could build applications such as an AI recycling bin that integrates rule-based AI and sensor technology (Ng et al., 2024a). Students could also learn to control Amazon Alexa using audio inputs by first writing a conversational program in App Inventor and then deploying it through the Alexa Developer Console (Hsu et al., 2022). To support students' foundational understanding of AI concepts, using non-programming activities through online tools, such as Machine Learning for Kids, could enable them to train AI models with minimal hardware requirements (Williams et al., 2023).

Middle school students could engage in machine learning projects that help them develop technical skills that may shape their attitudes toward AI (Kim & Lee, 2024). Students could use existing AI platforms (e.g., PoseBlocks) for real-time emotion and image recognition directly in web browsers (Williams et al., 2023). Other pre-trained AI models, such as those available in CAN Play and Text Generation tools, could be used to support students in creating AI-generated text and images while exploring deep learning concepts without requiring high-end computing power (Williams et al., 2023). By building their own object classification models, students gain hands-on experience with key machine learning steps, including data collection, training, and testing (Aung et al., 2022). To develop a well-rounded AI literacy, middle school students should be encouraged to integrate both hardware and software elements when creating AI prototypes such as Teachable Machine or Scratch AI (Ng et al., 2024a). Exposure to AI technologies relevant to machine learning and autonomous robotics, including speech and image recognition, can empower students to learn about AI systems in more depth (Williams et al., 2023).

By *high school*, the focus of this component evolves further, emphasizing student knowledge of creating AI-driven solutions for real-life scenarios in a professional field. High school students should engage in authentic, hands-on projects that bridge

their AI understanding with real-world applications. By developing AI systems and artifacts, students can apply AI concepts and techniques to solve meaningful problems across various domains. For example, students could create classification models to analyze large datasets, such as evaluating online customer reviews for restaurants, which is an approach commonly used in fields like marketing and customer service (Jiang et al., 2023a, 2023b). Additionally, students should have opportunities to design and develop AI-powered applications, including smartphone apps, autonomous robotics, and intelligent home assistants (Bellás et al., 2023; Kong et al., 2023; Lin et al., 2024). These experiences reinforce AI literacy by integrating technical knowledge with practical implementation.

To create functional AI systems, high school students should utilize a range of computing tools and programming languages, such as Python, Blockly, and industry-standard platforms. Moreover, students could prototype AI models using non-programming tools, such as Teachable Machine and StoryQ, before advancing to more complex implementations. For instance, students might develop predictive models to assess future water pollution levels in their communities, applying AI-driven data analysis techniques (Tena-Meza et al., 2022). Engaging in meaningful project-based learning allows students to develop programs embedded with key AI concepts while refining their computational thinking and programming skills, such as coding robotics with Python-based languages (Ng et al., 2024b; Sing et al., 2022). Through these experiences, students deepen their understanding of AI principles while exploring its transformative potential in solving real-world challenges.

Component 3: Interacting With AI Agents

For *elementary school* students, AI literacy competencies related to interacting with AI agents encompass several key areas ranging from observing AI agents' behaviors to assessing how these agents function and respond to various inputs (Yetişensoy & Rapoport, 2023). Specifically, research on *K–3* AI literacy and education has primarily focused on helping students grasp basic AI understanding by interacting with and observing the behaviors of AI agents, predominantly robots (Dai et al., 2024b). Students engage in these interactions by giving voice commands and using the robots for creative tasks such as storytelling or teaching the robots how to dance. Students in *grades 4–6* are expected to recognize various forms of AI agents besides robots, such as digital voice assistants like Siri and Alexa, robotic vacuums, self-driving cars, and chatbots like ChatGPT (Ottenbreit-Leftwich et al., 2023). Through such interaction, students in grades 4–6 should develop an improved understanding of the working principles behind such agents and expand their narrow understanding of supervised machine learning to deep learning and natural language processing (Ng et al., 2024b).

As AI becomes increasingly embedded in everyday life, *middle school* students should develop a clear understanding of AI-based tools and their applications. For instance, students should be able to explain the characteristics and general development process of AI technologies, as well as recognize their role in automation and system design (Park & Kwon, 2024). Students should be able to identify AI-integrated technologies and systems such as video recommendation systems (e.g.,

YouTube) or web-based searching engines (e.g., Google Search) which are different from non-AI systems such as video calls, game consoles, and wireless printers (Williams et al., 2023). Everyday AI applications such as fingerprint unlocking, face recognition, and voice translation serve as entry points for students to explore AI's real-world applications. Moreover, students could interact with intelligent agents like Amazon Alexa and program these agents using visual block-based coding platforms derived from Scratch Open Source Blocks (Druga et al., 2019). Activities like the "YouTube Scavenger Hunt" could challenge students to analyze AI-driven recommendation systems by identifying the predictions these systems make and the datasets they rely on (Eguchi et al., 2021). These experiences help students move from passive use of AI to active engagement, fostering critical thinking about how AI systems operate and make decisions.

By leveraging AI-powered tools in meaningful contexts, students can explore AI's potential beyond daily convenience, applying it to problem-solving in fields such as environmental protection (Kim & Lee, 2024; Ng et al., 2024a, 2024b). To develop a responsible and ethical understanding of AI, students should engage with AI tools that emphasize human-AI collaboration. Playful simulations, such as Quick, Draw! and Teachable Machine, provide opportunities for students to explore machine learning concepts interactively (Ng et al., 2024a). More advanced platforms like Machine Learning for Kids allow students to experiment with AI training models, helping them recognize the importance of data quality and quantity in AI decision-making (Kim et al., 2023). For instance, students could create AI-driven tic-tac-toe games using machine learning principles, reinforcing their understanding of AI's learning process (Kim et al., 2023). By actively designing and testing AI systems, students gain a deeper appreciation for the ethical considerations of AI, ensuring that AI is developed and applied in ways that align with human values and societal needs.

For *high school* students, AI literacy competencies related to interacting with AI agents emphasize a more critical and analytical approach, encouraging students to assess both the capabilities and limitations of these technologies. High school students frequently engage with a variety of AI agents, from facial recognition systems to chatbots and predictive analytics tools, prompting deeper inquiry into the mechanisms underlying their functionalities. Rather than simply interacting with AI, students at this level should critically examine instances where AI systems fail, such as inaccuracies in facial recognition, biases in chatbot responses, and errors in algorithmic predictions (Chiu et al., 2022). Through such interactions, students should not only recognize these failures but also explore the ethical and technical reasons behind them. Additionally, high school students should develop an awareness of how to strategically leverage AI tools to support and enhance their learning, such as using AI-powered writing assistants to refine their work or employing machine learning models for data analysis in STEM subjects (Utami et al., 2023). By engaging with AI agents in this way, students move beyond passive use and develop a nuanced understanding of how AI can be both a powerful tool and a source of challenges in various domains.

Component 4: Developing Awareness of AI Ethics

Students in *elementary school* should cultivate an emerging understanding of AI ethics related to recognizing the extent of AI's intelligence and truthfulness (Yang et al., 2024). Such understanding is fundamental for students to recognize the ethical implications (e.g., algorithmic bias) of AI technologies in various contexts. This, in turn, is essential for students to discern the boundaries of AI knowledge and develop strategies and solutions to mitigate these biases. Research has shown that students in *grades K–3* should be ready to develop emerging competencies related to awareness of AI ethics. Instead of explicitly articulating the attributes of AI ethics, students should be able to respond to questions related to how smart, friendly, and truthful they think AI agents are and recognize that AI can exhibit bias that stems from the representations of its training data (Chai et al., 2021). Students in *grades 4–6* are expected to develop skills to critically evaluate the capabilities, limitations, and ethical implications of AI. They should understand what AI knows and does not know and use this understanding to critically inform their decision-making with results from AI. Students in grades 4–6 should be able to understand both the positive and negative implications of AI (Ottenbreit-Leftwich et al., 2023). They are also expected to engage in discussions with peers and adults about advanced concepts, such as AI ethics and the biases that AI agents may exhibit (Kim et al., 2023). Students in these grades should also be able to articulate ideas about the broader societal roles of AI, such as effectively communicating these ideas to their peers (Coenraad, 2022).

As AI ethics gain prominence in educational contexts, *middle school* students may not be fully familiar with critical ethical considerations such as bias, privacy, transparency, and fairness (Kim et al., 2023). To foster ethical awareness, students should connect these concerns to foundational AI knowledge and how AI operates to better understand its ethical implications. It is crucial for middle school students to recognize that AI can introduce challenges related to privacy violations and biased decision-making, given its complex interactions between humans, machines, and environments (Kim et al., 2023). Developing this awareness requires a deep understanding of AI's characteristics and development process (Park & Kwon, 2024). To reinforce these concepts, students should engage in activities such as completing and presenting ethical matrices, where they identify key interest holders and analyze the potential positive and negative consequences of deploying AI projects (Williams et al., 2023).

When designing AI systems, middle school students need to understand the importance of integrating these ethical values by ensuring the reliability of AI models, maintaining transparency in user interactions, respecting privacy, and minimizing data bias (Ng et al., 2024b). To prepare students for the evolving AI landscape, it is essential to encourage discussions on technology equity and the responsible use of AI. Students should be able to identify and address bias in machine learning while considering the long-term impact of AI on their future careers and society (Zhang et al., 2023). Through in-class assignments, students should analyze key challenges in AI, including issues like algorithm bias and generalization (Aung et al., 2022). A significant focus in AI literacy is understanding why and how AI-related

technologies should be applied ethically, including topics such as data privacy, data protection, transparency, misinformation, and intellectual property rights (Aung et al., 2022). Students could also brainstorm ways in which technology can have both positive and negative societal effects (Coenraad, 2022). These discussions will equip students with the knowledge and critical thinking skills necessary to navigate ethical dilemmas in AI responsibly.

High school students should cultivate a sophisticated understanding of AI ethics by critically examining bias and ethical concerns that arise during the development and deployment of AI models. As students gain firsthand experience in designing, training, and implementing AI systems across various domains, they should actively analyze how bias can emerge in data and algorithms and explore strategies to mitigate these biases (Jiang et al., 2022, 2023b; Kong et al., 2023). Engaging in AI-driven projects allows students to integrate ethical considerations into their work, fostering a deeper awareness of the societal impacts of AI technologies (Kong et al., 2023). Further, high school students should be able to articulate and justify ethical decisions in AI design; engage in discussions about fairness, accountability, and transparency; and propose solutions that address ethical challenges in real-world AI applications.

Component 5: Understanding Human-AI Relationships

Students in *elementary school* should be empowered to learn and use AI confidently with strong AI readiness and self-efficacy. Most importantly, students should understand that AI is fundamentally created and guided by human decisions. With a well-rounded understanding of AI–human relationships, students should be able to envision, explore, and create approaches to leverage AI for generating innovative solutions. Students in *grades K–3* are expected to understand how AI can assist and improve people’s lives (e.g., the role and functions of robots in helping people). They should be able to identify its positive aspects and its potential to enhance daily life (Druga et al., 2019). Students in *grades 4–6* should comprehend the critical role humans play in developing AI and the ethical implications and responsibilities associated with AI technologies. This understanding should enhance their awareness of how AI is built, programmed, and operated. Such awareness encompasses the values and critical thinking skills needed to use AI responsibly and understand its impact on society. Empowered by an expanding understanding of AI, students should exhibit a more developed readiness and self-efficacy for using and learning about AI (Cai, 2024; Chai et al., 2021; Dai et al., 2024a).

Middle school students should engage in exploring societal problems to initiate AI-driven projects and propose solutions, fostering a deeper understanding of how humans play a fundamental role in designing and developing AI systems. Through these reflective experiences, students develop an awareness of the human decisions embedded in AI technologies. Additionally, students enhance their critical thinking by learning about GANs and their impact on daily life (Lee et al., 2021; Zhang et al., 2023). For example, through activities such as “GANs or Not,” students practice distinguishing between AI-generated and human-created media, strengthening their

ability to recognize algorithmic influences in modern digital content (Lee et al., 2021).

Middle school students should actively participate in discussions and activities that explore the human–AI relationship in real-world contexts. By presenting and evaluating AI-driven solutions among peers, students refine their understanding of AI’s societal impact and ethical considerations (Ng et al., 2024a). Hands-on experiences, such as examining AI-generated media, help students consider how AI influences social networks, misinformation, and media integrity (Zhang et al., 2023). Establishing connections between AI and students’ everyday experiences, such as through locally relevant AI applications like subway system chatbots, enhances their understanding of AI’s role in both local and global contexts. Additionally, engaging in structured debates, such as a Socratic seminar on the YouTube Kids app, encourages students to analyze the influence of AI on various interest holders and digital platforms (Druga et al., 2019). Exposure to real-world AI applications through case studies and problem-solving tasks—such as analyzing failures in facial recognition, chatbots, and AI-driven predictions—provides valuable insights into AI’s strengths and limitations (Park & Kwon, 2024). Additionally, students should investigate algorithmic bias, its societal and ethical implications, and ways to mitigate these biases. By exploring AI-related careers and recognizing their strengths, students can better understand the importance of continuous learning and adaptability in an AI-driven job market (Druga et al., 2019).

High school students should develop a comprehensive understanding of the impact of AI on society and critically reflect on the ethical responsibilities associated with AI development and deployment. They should recognize how AI operates at various social levels, bringing both opportunities and challenges that require thoughtful consideration. Students should explore how AI-driven systems function, including their motivations, limitations, and the role of human oversight in ensuring AI aligns with ethical and societal values (Bellás et al., 2023). With a strong foundation in AI knowledge and skills, students should be equipped to engage in meaningful discussions about leveraging AI for social good and designing AI-driven solutions that address real-world challenges in their communities (Xia et al., 2022). Additionally, students should develop proficiency in various AI methodologies, gaining the ability to apply AI tools effectively, interpret results, and integrate domain-specific knowledge when tackling complex problems (Tena-Meza et al., 2022). Through these experiences, students cultivate the confidence and capability to navigate and shape the evolving AI landscape responsibly and innovatively.

Discussion, Implications, and Conclusion

In this study, we reviewed and synthesized research on K–12 AI literacy and education, identifying more comprehensive and detailed components of AI literacy by specifying competencies for each grade band. We identified five key components of AI literacy that contribute to understanding how to help K–12 students become AI-literate. These components are (1) Understanding Foundational AI Concepts, (2) Creating AI Artifacts, (3) Interacting With AI Agents, (4) Developing Awareness

of AI Ethics, and (5) Understanding Human-AI Relationships. Our findings further specify the AI literacy components proposed in previous studies (Long & Magerko, 2020; Ng et al., 2021; Touretzky et al., 2019). For instance, our study extends the framework proposed by Ng et al. (2021) and that of UNESCO (Miao & Shiohira, 2024) by incorporating the component of Understanding Human-AI Relationships. Although many studies have emphasized the importance of this component in developing K–12 students' AI literacy (e.g., Eguchi et al., 2021), we argue that this component is crucial as it integrates the other four components (see Fig. 3), helping students understand human roles in developing, evaluating, and leveraging AI systems for social good.

Importantly, while our findings share some similarities with UNESCO's AI literacy framework (Miao & Shiohira, 2024), particularly in promoting ethical awareness, technical understanding, and active engagement with AI, there are important distinctions. UNESCO's framework is organized around four overarching dimensions and emphasizes developmental progression across three levels: understand, apply, and create. In contrast, our five-component framework emerged inductively from a synthesis of empirical studies and focuses on K–12 learners' competencies across grade bands. While UNESCO offers a high-level vision of AI literacy, our framework extends this to some extent by providing more granular, developmentally informed guidance for implementing AI literacy education in K–12 settings. For example, our component, "Understanding Human–AI Relationships," is not explicitly foregrounded in the UNESCO framework but is essential in our model, as it integrates cognitive, ethical, and relational aspects of AI interaction, which are highlighted as critical in many K–12 studies. Additionally, the five components identified in this study reflect a synthesis of previous AI literacy and education frameworks, reinforcing the importance of noncognitive competencies, particularly in early grades. At this stage, AI literacy efforts should focus on developing students' awareness, interest, and attitudes toward AI with minimal conceptual learning (e.g., Kim & Lee, 2024; Yang et al., 2024). This aligns with STEM education literature, which emphasizes the importance of sparking students' interest to enhance engagement and persistence in learning more cognitively demanding concepts in later grades (e.g., Maltese & Tai, 2010; Simoncini & Lasen, 2018; Wan et al., 2021). Thus, our findings suggest that AI education in elementary school should prioritize engaging students in enjoyable AI-related activities rather than focusing heavily on AI concepts.

Once students develop an interest in AI and grasp basic concepts, middle school education should transition to applying this knowledge to evaluate AI, as students gain access to mobile devices and social media. At this stage, they should also be encouraged to use AI tools to create solutions for personally meaningful and socially beneficial problems, engaging in problem-solving activities that involve collaboration with AI tools (e.g., Williams et al., 2023). In high school, students should develop a more in-depth understanding of AI, either broadly or within specific AI branches, with a focus on leveraging AI solutions in contexts relevant to their future workforce and careers (e.g., Jiang et al., 2023b). These insights can inform AI education researchers and curriculum developers in designing activities that engage K–12 students and develop their AI literacy effectively.

While many believe that AI literacy in K–12 should primarily be taught within technology, computer science, or AI courses, we argue that AI education should not be confined to these contexts (e.g., Yetişensoy & Rapoport, 2023). AI connects with various disciplines, and efforts should be made to integrate AI learning into other subjects rather than treating it as a stand-alone topic (Yang et al., 2025). The five AI literacy components and their associated competencies identified in this study can be considered subject-agnostic, making them versatile for integration into various subjects. This interdisciplinary approach would provide students with more opportunities to learn about AI and enhance their AI literacy (Yang et al., 2025). Emerging research has shown the connections among one or more of these AI literacy components being integrated into different subjects within STEM education. AI ethics and AI societal impact continue to be focal competencies for future STEM professionals, particularly in the context of responsible innovation in the digital age (Smith-Mutegi et al., 2025; Usher & Barak, 2024).

We believe that our synthesized AI literacy components and grade-band-specific competencies represent a critical first step toward developing a comprehensive learning progression framework for K–12 AI literacy and education. This could ultimately lead to the establishment of evidence-based K–12 AI education standards. However, achieving this goal requires further research, as our review included only 36 studies, which is insufficient for constructing a robust, stable, and generalizable AI literacy framework. Thus, we encourage AI literacy and education researchers to conduct further investigations into how and what students learn best about AI at each grade level. Researchers can also revalidate the findings from this study by developing continuous curricular activities from one grade band to another to examine whether the findings of our synthesis hold. Additionally, researchers should explore the challenges students face when learning certain AI concepts or using AI tools, which would enable a more comprehensive understanding of students' AI literacy development. Such an understanding is vital to supporting students in developing their AI literacy, as teachers can use knowledge about students' challenges to tailor instruction as well as provide more scaffolding for students.

Finally, it is equally important to consider the implications for teachers. Such knowledge, along with our framework, serves as a valuable foundation for developing professional development (PD) programs for educators who will be teaching AI. Our framework can help PD developers design more targeted PD programs based on grade levels, enabling teachers to focus on the competencies their students need to develop as well as the competencies teachers themselves require to effectively support AI literacy development. To this end, future research should explore how to structure teacher PD to be more meaningful and impactful, as well as investigate effective pedagogical approaches for teaching AI. The ultimate goal should be to support students' AI literacy development by equipping educators with the knowledge, skills, and strategies needed to facilitate AI education effectively.

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Declarations

Ethics Approval Not applicable.

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